

Voltage Support Capability of Multi-Hub Vehicle-to-Grid Operation Under Fleet Energy Constraints

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Abstract—Vehicle-to-grid (V2G) operation is often proposed as a voltage support resource on weak distribution feeders. Recent studies compare learned controllers against local droop control, but they do not establish how much support the connected fleet can actually provide. This paper computes an upper bound on that support. For every hour of a daily load profile, the active and reactive injection of each hub is optimized directly on the AC power flow, limited only by inverter rating. The bound is then used to separate two effects: the shortfall caused by limited stored energy, and the shortfall caused by the controller. On the IEEE 34-bus feeder, coordinated dispatch across five hubs clears every violating hour at a mild peak, while a single hub of the same rating clears fewer than half. Under realistic availability and state-of-charge limits, however, the fleet supplies less than 15 percent of the energy the controller requests. Energy adequacy outweighs controller choice by more than a factor of 40 at both load levels. Training the policy five times longer does not change this, and its commanded setpoints show why: it scales a nearly fixed power factor instead of allocating between active and reactive power as loading changes. A degradation-aware objective then removes about 60 percent of battery throughput for roughly a 1 percent rise in violation, because the marginal stored kilowatt-hour is nearly worthless once the fleet binds. Fleet sizing and availability therefore deserve more attention than controller sophistication.

Index Terms—vehicle-to-grid, voltage regulation, distribution networks, reinforcement learning, battery degradation, energy adequacy

I. INTRODUCTION

Long rural distribution feeders were not built for the voltage profiles they must now hold. Peak demand pulls the far end of the circuit below acceptable limits, and the usual remedies, such as reconductoring and additional regulators, are slow and expensive. Bidirectional inverters installed with electric vehicle (EV) fleets offer another option. They sit near the load end of the feeder, where support is most effective, and their cost is borne by the fleet owner. Recent work applies reinforcement learning (RL) to coordinate several V2G hubs for voltage regulation on the IEEE 34-bus feeder [1].

A wider body of work applies RL to inverter dispatch for the same purpose. Recent contributions include graph-based architectures with explicit safety layers [2], active management

of EV charging to hold voltage in low-voltage networks [3], battery-to-grid injection from charging and swapping stations [4], impact analysis of optimal bidirectional charging under spatial-temporal constraints [5], and coordination of local EV flexibility with social impact considered alongside network limits [6]. EV stock continues to grow [7], and with it the capacity notionally available for such services. These build on earlier RL formulations for volt/var control, both safe off-policy [8] and multi-agent consensus [9]. Across this literature the controller is treated as the binding constraint, so that a better policy is expected to extract more support from the same hardware.

That assumption has not been tested. None of these studies computes what the fleet could deliver under an ideal controller, so a reported improvement cannot be placed on a scale. A policy that halves the violation of a droop baseline may have captured most of the available benefit, or very little of it, and the published numbers do not separate the two cases.

This paper computes the missing reference point. For each hour we solve for the per-hub active and reactive injection that minimizes a two-sided voltage violation metric, using the AC power flow and limiting only inverter rating. No energy constraint is applied, so the result is an upper bound set by the power electronics alone. Comparing it against droop and learned controllers, run with and without a realistic fleet model, splits the total shortfall into an energy part and a control part.

Two design choices make that split trustworthy. Voltages are evaluated per phase rather than as bus averages, because the standard that defines the limits is itself per phase and a three-phase average hides single-phase excursions on an unbalanced feeder. Controllers are then compared on common random numbers, so every controller sees the same availability draw, the same initial state of charge and the same load realization, and the differences between them are paired rather than compared as independent means. Without pairing, the controller gap reported here would sit inside the scenario-to-scenario noise.

The practical stake is where a utility or fleet operator

TABLE I
SIMULATION AND TRAINING PARAMETERS

Parameter	Value
Test feeder	IEEE 34-bus, 24.9 kV
Hub rating (each)	500 kW / 400 kVAr
Peak multiplier, mild / aggressive	1.5 / 3.0
Fleet availability	45–85%
Arrival / minimum state of charge	70% / 20%
Droop deadband	± 0.02 p.u.
Actor and critic hidden layers	2×256
Learning rate / batch size	3×10^{-4} / 256
Training budget	20,000 and 100,000 steps
Seeds $\times$ scenarios	3×5

should spend effort. If the controller binds, the return comes from better control software, which is cheap to deploy across an existing installation. If stored energy binds, the return comes from more vehicles, longer dwell times or contractual availability, none of which a control upgrade can supply. These lead to different investments, and the two cases are distinguishable before any controller is built.

The contributions of this paper are as follows:

- We compute an hourly upper bound on multi-hub V2G voltage support for the IEEE 34-bus feeder and use it to split the shortfall into energy and control parts.
- We measure the active and reactive trade-off per unit of stored energy rather than per unit of apparent power.
- We compare droop and learned control under realistic availability and state-of-charge limits, and sweep the training budget to five times its usual value.
- We report a degradation-aware dispatch frontier and three negative results.

The remainder of this paper is organized as follows. Section II describes the feeder, the fleet model, the metrics and the controllers. Section III computes the upper bound and splits the shortfall. Section IV reports controller performance, Section V the degradation frontier, Section VI three further observations, and Section VII concludes.

II. SYSTEM MODEL AND STUDY DESIGN

A. Feeder and Charging Hubs

All experiments use the IEEE 34-bus test feeder, a 24.9 kV rural circuit whose length makes it a standard stress case for voltage studies, solved in OpenDSS through its Python interface. The single-hub configuration places one 500 kW/400 kVAr hub at bus 890. The multi-hub configuration distributes five hubs of the same rating at buses 890, 844, 832, 830 and 860, following [1]. A daily load shape drives hours 06:00 through 23:00, scaled by a peak multiplier. Two operating points are used throughout: a mild peak ($\lambda_{\max} = 1.5$) and an aggressive peak ($\lambda_{\max} = 3.0$). Table I collects the feeder, fleet and training settings used throughout.

B. Baseline

The reference study does not state whether the feeder regulators are active, and the answer changes every number

TABLE II
IDENTIFYING THE REGULATOR SETTING FROM THE NO-V2G BASELINE

Setting	Mild		Aggressive	
	Mean	Min	Mean	Min
Reported [1]	0.963	0.907	0.883	0.807
Regulators off (this work)	0.951	0.903	0.860	0.800
Regulators active (this work)	1.012	0.983	0.911	0.831

that follows. Its reported no-V2G statistics identify the setting. Those statistics are computed from hourly mean bus voltages, so that convention is matched in Table II only; elsewhere voltages are evaluated per phase, for the reason given in Section II-D.

With the regulators disabled, our minimum voltages fall within 0.004 to 0.007 p.u. of the reported values. With them active, the mild minimum differs by 0.076 p.u., which is outside any plausible modelling difference. Our feeder mean runs 0.012 to 0.023 p.u. low, which we attribute to load allocation details not given in the reference, so the configuration is identified from the minimum rather than the mean. All later results use disabled regulators.

C. Fleet and Battery Model

Each hub serves a fleet whose availability varies through the day between 45 and 85 percent and is drawn independently per scenario. Vehicles arrive at 70 percent state of charge and are not discharged below 20 percent. State of charge is drawn down by apparent power, so reactive support costs the same stored energy as active support. This follows the reference formulation and matters for Section III-C.

The distinction between an unconstrained and a constrained fleet runs through the whole study. The reference work evaluates multi-hub coordination while “assuming sufficient EV capacity at each hub” [1]. We report both cases, because the difference between them turns out to be the principal result.

D. Evaluation Metrics

ANSI C84.1 [10] is a per-phase standard, so voltages are evaluated per phase rather than as bus averages, which mask single-phase excursions on this unbalanced feeder. The primary metric is the day-integrated two-sided violation

$$IV = \sum_h \sum_p \left[\max(0, v_{\min} - v_{p,h}) + \max(0, v_{p,h} - v_{\max}) \right] \Delta t \quad (1)$$

in per-unit-hours, with $v_{\min} = 0.95$, $v_{\max} = 1.05$ and $\Delta t = 1$ h. Unlike a count of violation hours, (1) does not saturate when two controllers violate in the same hours by different margins, and being two-sided it penalizes overvoltage as well.

All comparisons use common random numbers. Every controller sees the same availability draw, initial state of charge and load realization, and differences are paired. Confidence intervals are 95 percent over the paired scenarios and training seeds.

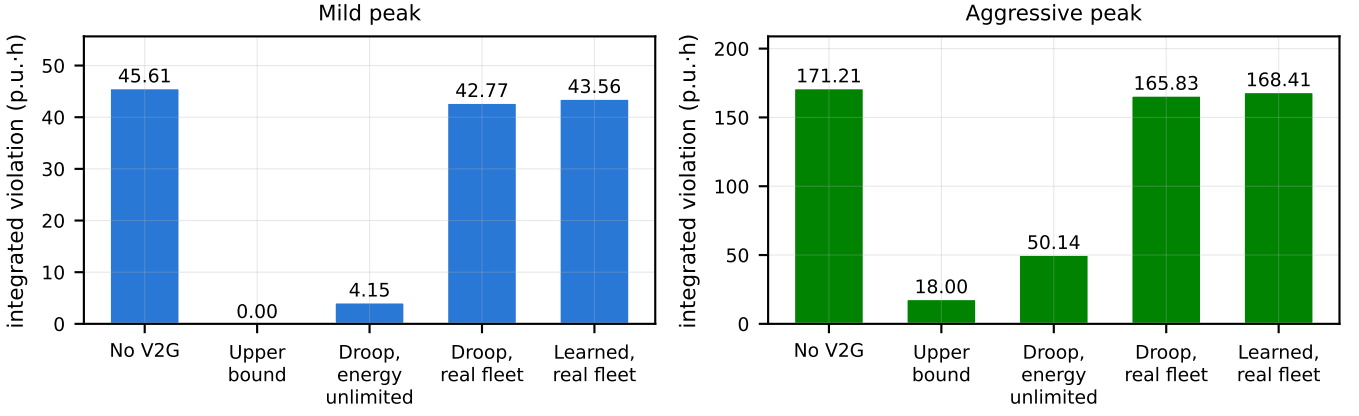


Fig. 1. Day-integrated voltage violation on the five-hub feeder.

E. Control Algorithms

The droop baseline is a Volt-Watt and Volt-VAr characteristic with a ± 0.02 p.u. deadband, consistent with IEEE 1547 [11]. Droop response and feeder voltage are mutually dependent, so the baseline is not evaluated open-loop. We obtain the fixed point by damped Jacobi updates with a residual convergence test, a detail examined in Section VI.

The learned controller is soft actor-critic [12] as implemented in Stable-Baselines3 [13], with two hidden layers of 256 units, a learning rate of 3×10^{-4} and a batch size of 256. Episodes span one day, so that rationing of a finite battery over time is representable. The state comprises bus voltages, load multiplier, hour of day, and per-hub state of charge and availability. The action sets per-hub active and reactive power as a residual on the droop prior.

III. VOLTAGE SUPPORT CAPABILITY

A. Upper Bound on Voltage Support

For each hour we search for the per-hub injection (P_i, Q_i) that minimizes (1) subject to $\sqrt{P_i^2 + Q_i^2} \leq S_i^{\text{rated}}$, by coordinate descent over a refined polar grid with a full AC power flow at every candidate and no energy constraint imposed. Non-converged solutions are masked rather than read, because at deep-sag hours the high-injection end of the search lies past the nose of the power-voltage curve.

Searching per hub rather than uniformly is necessary. With five hubs at equal output, raising the feeder end to 0.95 p.u. drives the near buses above 1.05 p.u., so hours an uneven dispatch clears comfortably are scored unclearable, and the five-hub case then scores worse than the single-hub case, which is not physically possible. Per-hub search removes the artefact and moves the mild multi-hub result from five clearable hours to eighteen.

Table III reports the outcome. A single hub at full rating clears seven of the eighteen loaded hours at the mild peak and two at the aggressive peak. Five hubs of the same individual rating, dispatched unevenly, clear all eighteen and four respectively. Fig. 2 shows the hourly picture. The reachable

TABLE III
VOLTAGE SUPPORT UPPER BOUND UNDER OPTIMIZED PER-HUB DISPATCH

Configuration	Clearable hours (of 18)	Day-total IV (p.u.·h)	
		Uniform	Optimized
Single hub, mild	7	11.14	10.78
Single hub, aggressive	2	125.84	123.97
Five hubs, mild	18	0.29	0.00
Five hubs, aggressive	4	28.33	18.00

envelope sits on or above the 0.95 p.u. limit for the entire mild day, and falls short of it for eleven hours of the aggressive day regardless of dispatch. Those eleven hours are a siting and rating problem, not a control problem, and no policy can clear them. At equal total inverter rating, spreading capacity across five hubs and dispatching it unevenly reduces violation by 100 percent at the mild peak and by 36.5 percent at the aggressive peak, relative to uniform dispatch.

B. Energy and Control Contributions

Fig. 1 places the controllers against this bound and is the central result of the paper. Reading the mild panel, the untreated feeder accumulates 45.61 p.u.·h of violation and the bound removes all of it. A droop law with unlimited stored energy removes 41.46, or 91 percent, while the same law with a realistic fleet removes 2.84, or 6 percent. The fleet delivers 1342 kWh against the 9264 kWh the unconstrained law requests, a ratio of 14.5 percent. At the aggressive peak the ratio is 5.7 percent.

The controller span lies inside that shortfall. Between tuned droop and the best learned policy we obtained, the difference is 0.79 p.u.·h at the mild peak and 2.58 at the aggressive peak, against energy shortfalls of 38.62 and 115.69. Energy adequacy therefore outweighs controller choice by a factor of 49 and 45.

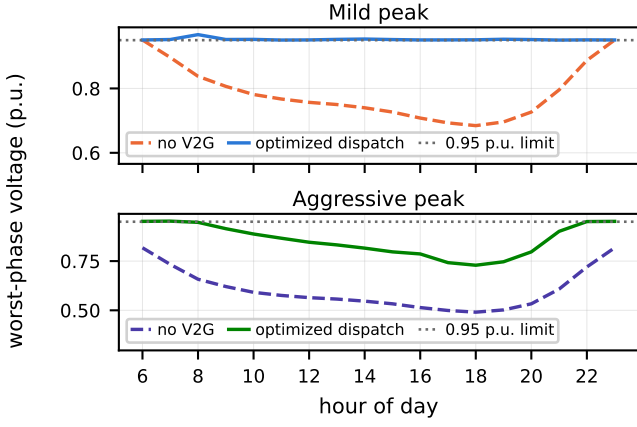


Fig. 2. Worst-phase voltage by hour, with and without optimized five-hub dispatch.

TABLE IV
FIVE-HUB FEEDER WITH REALISTIC FLEET CONSTRAINTS

Controller	Mild		Aggressive	
	IV (p.u.·h)	Throughput (kWh)	IV (p.u.·h)	Throughput (kWh)
No V2G	45.61	0	171.21	0
Droop, real fleet	42.77	1342	165.83	1582
Learned, 20k steps	45.45	5842	169.60	3227
Learned, 100k steps	43.56	5588	168.41	3176
Droop, unlimited	4.15	9264	50.14	27814

C. Active and Reactive Trade-off

Because state of charge is drawn down by apparent power, the common assumption that reactive power is free does not hold. Sweeping the injection angle at fixed apparent power and fixed loading, we find that active power delivers between 1.25 and 1.49 times the worst-phase voltage rise of reactive power for the same battery energy withdrawn. The voltage-optimal split is 35 to 40 degrees from the active axis at half rating, and moves towards 17 to 26 degrees at full rating as the resistive voltage drop becomes dominant. The hub rating ratio of 400 kVAr to 500 kW corresponds to 38.7 degrees.

This carries a design consequence. On a feeder with this R/X character, a hub sized by apparent power and operated at its rating ratio commits more reactive power than is useful, and pays full battery energy for it.

IV. CONTROLLER PERFORMANCE UNDER FLEET CONSTRAINTS

A. Multi-Hub Operation

Table IV reports the five-hub feeder with the fleet model active, over three seeds and five paired scenarios. Droop reaches 42.77 p.u.·h at the mild peak and 165.83 at the aggressive peak. The learned policy trained on the conventional 20,000-step budget reaches 45.45 and 169.60, and loses to droop in every paired scenario at both load levels.

Two features of the table are worth noting. The learned policy consumes 5842 kWh of battery throughput at the mild peak against 1342 kWh for droop, and converts that into a worse voltage outcome. At the mild peak its result is statistically indistinguishable from installing no V2G at all. Both controllers respect the upper voltage limit exactly, at 1.050 p.u., a point taken up in Section VI.

B. Training Analysis

A negative result about a learned controller invites the objection that it was not trained long enough, and at 20,000 steps, about 1100 episodes for a ten-dimensional action space, the objection is reasonable. We therefore trained to 100,000 steps and evaluated at 20,000-step checkpoints on identical scenarios, placing checkpoints on the chunk boundaries the fixed-budget runs use so the 20,000-step row measures the agent of Table IV.

The outcome differs by load level, as Fig. 4 shows. At the aggressive peak the curve is flat from 40,000 steps onward. Violation moves from 169.20 to 168.41 across a fivefold increase in budget, which lies inside the combined confidence interval of 1.13, and droop wins every paired scenario at every checkpoint. The advantage of droop under aggressive stress is therefore not an artefact of training budget.

At the mild peak the curve is still improving at 100,000 steps. Violation falls from 45.05 to 43.56, a change of -1.48 against a confidence interval of 0.49. The gap to droop narrows monotonically from $+2.27$ to $+0.79$, and the policy takes two of ten paired scenarios, the only scenarios it wins anywhere in this study. We therefore report the mild comparison as unresolved at the budgets tested rather than as a droop victory. Set against the previous section, however, the gap narrowed by 1.48 p.u.·h over a fivefold increase in training, against an energy shortfall of 38.62 at the same operating point.

C. Allocation Analysis

A scoreboard alone does not explain the outcome, so we examined the active and reactive split the policy commands, hour by hour, weighted by apparent power and compared against the voltage-optimal angle recomputed at the injection level the policy itself selected.

The policy does not track the optimum, and the deviation does not fall with more training. At 20,000 steps the mean absolute deviation is 26.5 degrees at the mild peak and 32.5 degrees at the aggressive peak. At 100,000 steps these become 20.8 and 31.5 degrees. The changes of -5.7 and -1.0 degrees both lie inside their combined 95 percent confidence intervals of 24.2 and 35.1 degrees, so neither is a reduction the data supports.

Fig. 3 shows the hourly picture at 100,000 steps. The voltage-optimal angle stays between 29 and 35 degrees for most of the day at both load levels. At the mild peak the commanded angle stays near 20 degrees, with an apparent-power-weighted daily mean of 18.8 degrees. At the aggressive peak the daily mean falls to 1.2 degrees, which is almost pure active power, and in 50 percent of sampled hub-hours the

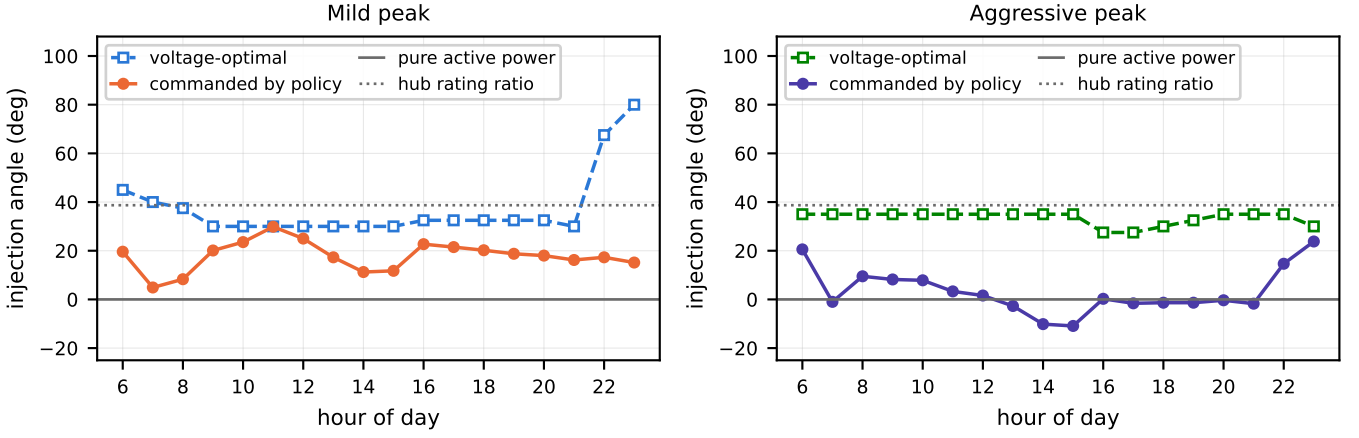


Fig. 3. Commanded injection angle against the voltage-optimal angle, by hour.

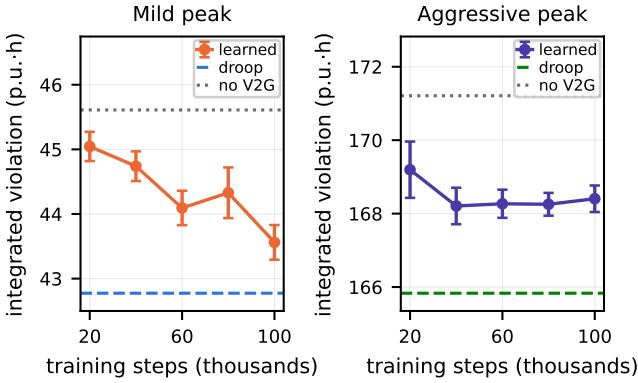


Fig. 4. Violation against training budget.

commanded reactive power is negative. Reactive absorption lowers voltage on this feeder, so these are actions of the wrong sign for the condition being corrected, not merely conservative ones. The pattern is consistent with a controller that scales a roughly fixed power factor instead of allocating between active and reactive power in response to loading.

V. DEGRADATION-AWARE DISPATCH

Battery wear is the standing objection to V2G participation, and the economic case is sensitive to it [14]. The reference formulation carries no battery term in its reward. We added an amp-hour throughput cost, expressed on the same scale as the voltage penalty, and swept its weight across three orders of magnitude.

The resulting frontier is flat, as Fig. 5 shows. Across a 300-fold sweep the violation stays within a 2.6 percent band, between 44.83 and 46.01 per-unit-hours, while daily throughput falls from 6188 to 2449 kWh. Comparing the endpoints, roughly 60 percent of battery throughput can be removed for a 1.1 percent increase in violation.

This follows from the energy-limited regime identified earlier. When the fleet cannot supply what the feeder needs, the

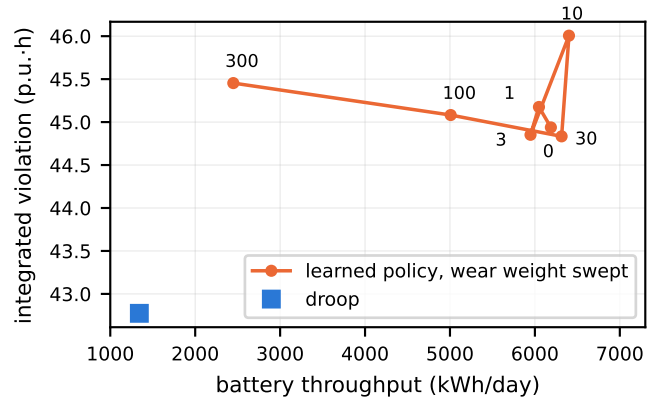


Fig. 5. Violation against battery throughput as the wear weight is swept from 0 to 300.

marginal voltage value of the last kilowatt-hour discharged is very low, so a wear penalty removes it at almost no cost. On feeders in this regime, aggressive degradation-aware dispatch therefore costs little in voltage performance, which improves the participation case for fleet owners.

The intermediate points should not be over-interpreted: each weight was trained with a single seed and the sequence is not monotone, so the defensible claims are the endpoint comparison and the width of the band, not the shape in between.

VI. ADDITIONAL OBSERVATIONS

This section reports three checks that did not produce the expected effect. Each addresses a design choice that is common in the literature.

A. Safety Projection

Projecting commanded setpoints onto the voltage-feasible set is standard practice in learning-based volt/var control [2], [8], so we implemented and measured two: a proportional

reduction along the commanded ray, and a reactive-first variant motivated by Section III-C.

Both are inert. The unprojected policy already reports zero overvoltage and a maximum phase voltage of exactly 1.050 p.u. at both load levels, and projection changes the violation metric by less than the confidence interval. The feasibility test acts on commanded setpoints, but the state-of-charge limit clips those commands first, so the energy constraint binds before the voltage constraint and leaves nothing to remove. Where overvoltage does appear, it tracks the training regime rather than the controller: policies trained on independently sampled load multipliers without a fleet in the loop reach 1.229 p.u., whereas day-structured training with the fleet in the loop holds 1.050 everywhere we tested.

B. Fleet State

Adding state of charge and availability to the observation, information the reference formulation does not provide, does not improve voltage performance. At the mild peak the augmented agent reaches 45.41 ± 0.36 against 44.91 ± 0.29 for a voltage-only observation, an overlapping interval. What it changes is energy use. The augmented agent spends 6068 kWh against 2497 kWh for no violation benefit, while the voltage-only agent breaches the upper limit at the aggressive peak where the augmented agent does not.

C. Droop Convergence

The last observation is methodological and affects any study using a closed-loop droop baseline, whose fixed point is commonly computed with a fixed iteration count. On the five-hub feeder, at a damping factor of 0.6, the iteration does not settle but enters a period-two limit cycle, with total active power alternating between 1298 and 630 kW at the mild-peak evening hour. A fixed iteration count then returns whichever branch the last iteration reached, and the reported baseline flips with the parity of the count. Damping of 0.3 with a residual test converges in all four configurations tested, so a convergence diagnostic should accompany any droop baseline computed this way.

VII. CONCLUSION

We computed an upper bound on multi-hub V2G voltage support for the IEEE 34-bus feeder and used it to separate the shortfall due to control from the shortfall due to stored energy. Uneven coordination across five hubs is of real value, clearing every violating hour at a mild peak where a single hub of equal rating clears seven of eighteen. Once realistic availability and state-of-charge limits are applied, however, the fleet supplies well under a fifth of the energy the controller requests, and the resulting shortfall exceeds the entire span between a tuned droop law and a converged learned policy by a factor of roughly 45 to 50.

This is not an argument against learned control in distribution systems, but an argument that the operating regime should be established before the controller is designed, because in the energy-limited regime a better policy has very little left

to win. Both the upper bound and an energy adequacy ratio follow from feeder and fleet data before any controller exists, and we suggest reporting them as standard context alongside controller comparisons.

Two further results point the same way. A degradation-aware objective removes 60 percent of battery throughput for a 1.1 percent violation penalty, because the marginal kilowatt-hour is nearly worthless in this regime. A safety projection layer, standard elsewhere, is measurably inert here because the energy constraint binds first.

The study covers one feeder and one learning algorithm, and the mild-peak comparison remains unresolved at the budgets we could afford. The ratio reported here depends on feeder impedance and hub siting, so it should be recomputed on a larger network. More usefully, the bound is inexpensive enough to invert: rather than scoring a controller against it, it can serve as a planning screen returning the fleet size and availability needed to make a target violation level reachable.

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